

- 33
 66
 330
 330
- a) $T = 327.86^\circ\text{F}$ $V = 0.01774 \text{ ft}^3/\text{lb}$ $h = 298.6 \text{ BTU}$
 b) $P = 72.91 \text{ bar}$ $V = 0.02437 \text{ m}^3/\text{kg}$ $U = 327.13 \text{ kJ/kg}$
 c) $T = 295.06^\circ\text{C}$ $U = 2569.8 \text{ kJ/kg}$ $h = 2758 \text{ kJ/kg}$
 d) $P = 1.5299 \text{ bar}$ $V = 24.17 \text{ m}^3/\text{kg}$ $U = 0.71 \text{ m}^3/\text{kg}$

2. $T_1 = 180^\circ\text{C}$ $T_2 = 100^\circ\text{C}$ $1 \text{ bar} = 0.1 \text{ MPa}$
 a) $P = 1.002 \text{ MPa}$ 10.02 bar $(0.1) \text{ MPa}$
 b) $X = 0.1154$

$P_2 = 1.014 \text{ bar}$ $V_1 = 0.19405 \text{ m}^3/\text{kg} = V_2$
 $P_2 = 0.1014 \text{ MPa}$ $V_{f2} = 1.0435 \times 10^{-3} \text{ m}^3/\text{kg}$

$V_{g2} = 1.673 \text{ m}^3/\text{kg}$
 $V_2 = V_{f2} + x_2 (V_{g2} - V_{f2})$
 $x_2 = \frac{V_2 - V_{f2}}{V_{g2} - V_{f2}} = \frac{0.19405 - (1.0435 \times 10^{-3})}{1.673 - (1.0435 \times 10^{-3})}$
 $x = 0.1154$

3. $m = 2 \text{ kg}$ $P = 300 \text{ kPa} (3.0 \text{ bar})$

$V_2 = 2.064 \text{ m}^3$

a) $= 116.246^\circ\text{C}$ 60°C $?$ 3.0
 b) $= 370.83^\circ\text{C}$ 60°C 140°C 3.613
 c) 365.8°C 2.036

$W = P(V_2 - V_1)$

$300 (2.064 - 0.422)$ 3.613 2

$W = 492.6 \text{ kPa} \cdot \text{m}^3 = 492.6 \text{ kJ}$ $V_1 = 0.422$

66

4. Refrigerant 134a

$$P_1 = 10 \text{ bar} \quad P_2 = 2 \text{ bar} \quad m = 1 \text{ kg}$$

$$T_1 = 60^\circ\text{C} \quad T_2 = 30^\circ\text{C} \quad W = 20 \text{ kJ}$$

$$a) V_1 = 0.0114 \text{ m}^3/\text{kg} (1 \text{ kg}) = 0.0114 \text{ m}^3$$

$$b) Q = W + m(u_2 - u_1) \quad V_2 = 0.0265 \text{ m}^3$$

$$Q = 20 + 1(0.0114 - 0.0265)$$

$$Q = 19.9 \text{ kJ} \quad 80\%$$

5. $m = 1.50 \text{ kg}$

$$T_1 = 70^\circ\text{C} \quad T_2 = 120^\circ\text{C}$$

$$a) 0.3119 \text{ bar} (10^2) = 31.19 \text{ kPa}$$

$$b) \quad v_1 = 1.0228 \times 10^{-3}$$

$$v_g = 5.042$$

$$u_1 = 292.95$$

$$v_2 = 0.89133 - v_1$$

$$v_g = 2578.9$$

$$v_1 = v_f + x_1 v_{fg}$$

$$0.89133 = 0.0010228 + x_1 (5.042 - 0.0010228)$$

$$x_1 = 0.1766$$

$$292.95 + (0.1766)(2176.65)$$

$$u_1 = 677.346$$

$$Q = m(u_2 - u_1)$$

$$(1.5)(2328.9 - 677.346)$$

$$Q = 2477.3 \text{ kJ} \quad 80\%$$

6. Refrigerant 22

$$P = 10 \text{ bar} \quad m = 25 \text{ kg} \quad \text{quality} = 60\% = x = 0.6$$

$$v_f = 0.8352 \times 10^{-3} \quad v_g = 0.0236 \text{ m}^3/\text{kg}$$

$$v = v_f + x(v_g - v_f)$$

$$v = 0.00352 \times 10^{-3} + 0.6(0.0236 - 0.8352 \times 10^{-3})$$

$$v = 0.01449 \text{ m}^3/\text{kg}$$

$$v_g = 0.835 \text{ m}^3 / 37.5 \text{ kg} (0.0236)$$

$$v_1 = 25(0.8352 \times 10^{-3}) =$$

$$= 0.02088 \text{ m}^3$$

$$V = (25 \cdot 37.5)(0.01449)$$

$$V = 0.9656 \text{ m}^3$$

$$\frac{v_g}{v}$$

$$\frac{0.835 \times 100}{0.9056} \frac{\text{m}^3}{\text{m}^3} \times \frac{\text{m}^3}{\text{m}^3} = 0.6$$

$$0.6 = \frac{m_g}{m_g + 25}$$

$$m_g = 37.5 \text{ kg}$$



$$7. T_C = 5.2 \text{ k} \quad P_C = 2.3 \text{ bar} \quad M = 4.003 \frac{\text{g}}{\text{mol}}$$

$$T_1 = 27^\circ\text{C} = 300 \text{ K}$$

$$P_1 = 1.3 \text{ bar}$$

$$T_2 = 17^\circ\text{C} = 290 \text{ K}$$

$$P_2 = 0.8 \text{ bar}$$

$$R = \frac{R_u}{M} = \frac{8.314}{4.003} = 2.0769 \text{ J/mol}\cdot\text{K}$$

$$T_v = \frac{300}{5.2} = 57.69$$

$$k = \frac{1.3}{2.3} = 0.565$$

$$V_1 = \frac{m R T_1}{P_1}$$

$$V_2 = \frac{M k T_2}{P_2}$$

$$V_1 = \frac{(0.2)(10^{-3})(2.0769)(300)}{(1.3)(10^2)}$$

$$V_1 = 959 \text{ cm}^3$$

$$V_2 = \frac{(0.2 \cdot 10^{-3})(2.0769)(290)}{(0.8)(10^2)}$$

$$V_2 = 1506 \text{ cm}^3$$

$$6. T_1 = 25^\circ\text{C} = 298 \text{ K}$$

$$P_1 = 2.75 \text{ bar} \quad 2.75 \text{ kPa} \cdot V = m v T$$

$$V = 0.05 \text{ m}^3$$

$$T_2 = 257^\circ\text{C}$$

$$Q = 2.75 \text{ kJ}$$

$$R = \frac{8.314}{44.01}$$

$$= 0.1889$$

$$m = \frac{P V}{R T}$$

$$m = \frac{2.75(0.05)}{(0.1889)(298)}$$

$$m = 0.244 \text{ kg}$$

$$W = 80\%$$